Course One
Foundations of Data Science

Instructions

Use this PACE strategy document to record decisions and reflections as you work through this end-of-course project. You can use this document as a guide to consider your responses and reflections at different stages of the data analytical process. Additionally, the PACE strategy documents can be used as a resource when working on future projects.

Course Project Recap

Regardless of which track you have chosen to complete, your goals for this project are:

- Complete the PACE Strategy Document to plan your project while considering your audience members, teammates, key milestones, and overall project goal.
- Create a project proposal for the data team.

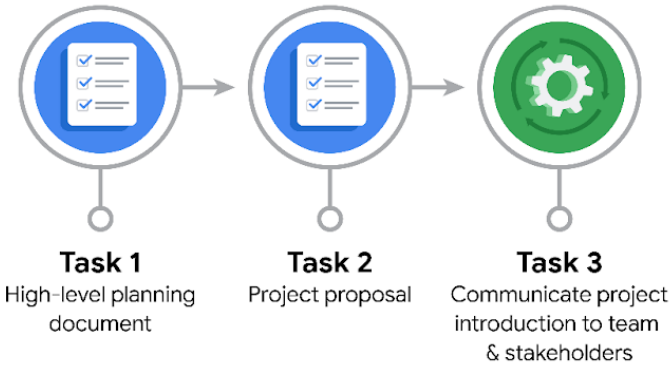
Relevant Interview Questions

Completing this end-of-course project will empower you to respond to the following interview topics:

- As a new member of a data analytics team, what steps could you take to get 'up to speed' with a current project? What steps would you take? Who would you like to meet with?
- How would you plan an analytics project?
- What steps would you take to translate a business question to an analytical solution?
- Why is actively managing data an important part of a data analytics team's responsibilities?
- What are some considerations you might need to be mindful of when reporting results?

Reference Guide

This project has three tasks; the following visual identifies how the stages of PACE are incorporated across those tasks.



Data Project Questions & Considerations

PACE: Plan Stage

- Who is your audience for this project?

The audience for this project is the New York City Taxi and Limousine Commission (TLC).

- What are you trying to solve or accomplish? And, what do you anticipate the impact of this work will be on the larger needs of the client?

I am trying to predict the taxi fares before the ride using a regression model and historical data. The impact of this work will be that TLC will be able to estimate the fare's of each journey before it happens.

- What questions need to be asked or answered?

How transparent is the model? How will the models predictions be used?

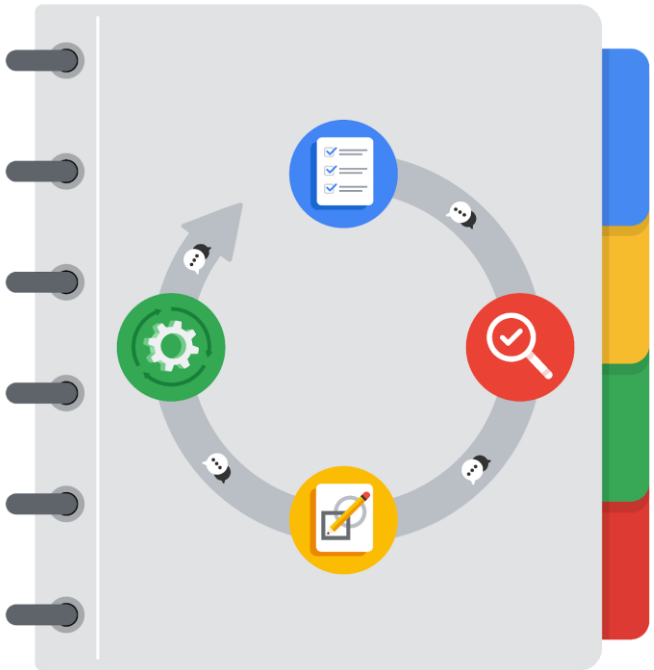
- What resources are required to complete this project?

Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn and SciPy.

- What are the deliverables that will need to be created over the course of this project?

Project Proposal, EDA Report, Hypothesis Test, Regression Model building and ML Model building.

THE PACE WORKFLOW



[Alt-text: The PACE Workflow with the four stages in a circle: plan, analyze, construct, and execute.]

You have been asked to demonstrate for the company's data team how you would use the PACE workflow to organize and classify tasks for the upcoming project. Select a PACE stage from the dropdown buttons. A few tasks involve more than one stage of the PACE workflow. Additionally, not every workplace scenario will require every task. Refer back to the Course 1 end-of-course portfolio project overview reading if you need more information about the tasks within the project.

Project tasks

Following are a group of tasks your company's data team has determined need to be completed within this project. The data analysis manager has asked you to organize these tasks in preparation for the project proposal document. First, identify which stage of the PACE workflow each task would best fit under using the drop down menu. Next, give an explanation of why you selected the stage for each task. Review the following readings to help guide your selections and explanation: The PACE stages and Communicate objectives with a project proposal. You will later reorder these tasks within a project proposal.

1. **Evaluating the model:** Construct

   Why did you select this stage for this task?

   I chose the Construct stage as when evaluating the models predictions you can do model selection and any fine tuning of the model. This therefore fits into the Construct stage of the PACE Workflow.

2. **Conduct hypothesis testing:** Analyze **and** Construct

   Why did you select these stages for this task?

   This is in the Analyze and Construct Stages as when conducting an Hypothesis Test the variables are not just being statistically analyzed but also a statistical test with a conclusion is being made.

3. **Begin exploring the data:** Analyze

   Why did you select this stage for this task?

   I chose the Analyze stage for this task as it is the initial exploration of the data and thereby fits into the Analyze stage.

4. **Data exploration and cleaning:** Plan **and** Analyze

   Why did you select these stages for this task?

   The Plan and Analyze stages were chosen for this task as data exploration and cleaning falls under the Analyze stage already but when dealing with missing data or outliers you have to plan on how to deal with it and this is where the task also falls into the Plan stage.

5. **Establish structure for project workflow (PACE):** Plan

   Why did you select this stage for this task?

   This task was chosen for the Plan stage as it is where the beginning of the project takes place and where the workflow is established and therefore fits into the Plan stage.

6. **Communicate final insights with stakeholders:** Execute

   Why did you select this stage for this task?

   This task was placed into the Execute stage as this is where the final results of the project are shared with the stakeholders and so is the final part of the project and so is in the Execute stage.

7. **Compute descriptive statistics:** Analyze

   Why did you select this stage for this task?

   This is in the Analyze stage because it is the initial analysis of the data.

8. **Visualization building:** Analyze **and** Construct

   Why did you select these stages for this task?

   This is in the Analyze and Construct stages of the PACE Workflow due to visualizations being built in EDA as well as dashboards being built for the stakeholders.

9. **Write a project proposal:** Plan

   Why did you select this stage for this task?

   This is because the task sets out the tasks, milestones and deliverables and so is in the planning stages of a project and therefore in the Plan stage of the PACE Workflow.

10. **Build a regression model:** Analyze **and** Construct

    Why did you select this stage for this task?

    The stages Analyze and Construct were selected for this task as when building a regression model your able to analyze the data via regression outputs as well as to construct a model, thereby fitting into both stages.

11. **Compile summary information about the data:** Analyze

Why did you select this stage for this task?

This was chosen for the Analyze stage as it is getting the initial information about the data and therefore fits into the Analyze stage.

12. **Build machine learning model:** Construct

Why did you select this stage for this task?

This was chosen for the Construct stage as this is where the ML model is built and fitted to the data and so is the construction of the model and therefore in the Construct stage.